

Publishing eLML Lessons from Lenya

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Chapter 1

Publishing eLML Lessons from Lenya

1.1 Prerequisites

1.1.1 Variables

We'll use the following variables:

- `$SVN` : shortcut for `https://idsubversion.uzh.ch/repos/unicms/trunk`
- `$PUBLISH_USECASE` : shortcut for `$SVN/modules/learning/elml/usecases/elml/publish`
- `$WEBAPP` : shortcut for the web application installation directory (e.g., `$TOMCAT_HOME/webapps/ROOT`)

1.2 Installation and Configuration

1.2.1 Quick Start

To install the eLML module,

- Edit `$PUBLISH_USECASE/ant/user.properties` ,
- Edit `$SVN/modules/learning/build.properties` ,
- Go to `$SVN/modules/learning` ,
- Execute the command `ant` , and
- (Re)Start your servlet engine.

The publish usecase requires Ant 1.7. At the moment, Lenya ships with Ant 1.6.5, so it has to be upgraded:

- Remove `$WEBAPP/WEB-INF/lib/ant*.jar`
- Copy `*.jar` from `$PUBLISH_USECASE/lib/ant` to `$WEBAPP/WEB-INF/lib`

1.3 The eLML Java Classes

1.3.1 The eLML API

To ensure reusability, the actual eLML functionality is isolated as an API, using the package `ch.elml`. This API should expose all features offered by eLML-related objects, i.e. it is an abstraction of the model layer. All client code should handle eLML operations only via this API. The API interfaces are located in the directory `$SVN/modules/learning/elml/java`. For further information on the single interfaces, please consult the JavaDocs.

1.3.2 The Lenya-specific eLML Implementation

The Lenya-specific implementation of the eLML API is located in the package `ch.unizh.lenya.elml`. The classes are located in the directory `$SVN/modules/learning/elml`. For each eLML API interface, you'll find a corresponding implementing class. The implementations make use of Apache Avalon and Excalibur mechanisms like configuration and source resolving. For more information, consult the Excalibur website.

The entry point to the eLML object model is provided by the `LenyaWorkspaceFactory`. The workspace factory is configured in `$WEBAPP/WEB-INF/cocoon.xconf`. The learning module build process patches `cocoon.xconf` using the patch file `$SVN/modules/learning/elml/config/workspace-factory.xconf`, setting the workspace location based on the `elml.workspace` property in `$SVN/modules/learning/`.

1.3.3 Lenya-specific eLML Usecases

The package `ch.unizh.lenya.elml.usecases` contains classes providing additional functionality for eLML-related objects in Lenya. For more information on the single classes, please consult the JavaDocs.

1.4 The Publish Usecase

1.4.1 The Publish Usecase

The publish usecase consists of the following parts:

- Controller (FlowScript, usecase sitemap) to control the user interaction
- View (JX templates) to collect user input
- Java classes to initialize and invoke the Ant process
- Ant buildfile implementing the actual publishing process
- JAR libraries containing classes used by the publishing process

All files, except the Java classes, are located in `$SVN/modules/learning/elml/usecases/elml/publish`.

1.4.2 The Usecase Sitemap

The usecase sitemap `$SVN/modules/learning/elml/usecase-publishlesson.xmap` calls the FlowScript functions and serves usecase-related requests (e.g. log files, ZIP files). It also provides the pipelines for the usecase view, based on JX templates.

1.4.3 The FlowScript

The FlowScript `$PUBLISH_USECASE/publish.js` takes care of the view page flow and user input validation. It doesn't contain any functionality to manipulate model objects directly.

1.5 The Ant Build Process

1.5.1 The Build File

The Ant build process is defined by the buildfile `$PUBLISH_USECASE/ant/build.xml`. To get more information on the targets, you can execute the following command in the `$PUBLISH_USECASE/ant` directory:

```
ant -projecthelp
```

1.5.2 Configuration

The directory `$PUBLISH_USECASE/ant` contains two configuration files:

- `user.properties` is used to configure the global publishing settings, e.g. the server addresses for deployment. It is meant to be edited.
- `params.properties` lists the parameters which are passed to the build process. This file is not meant to be edited. If you want to run the build process from the command line, copy this file to the eLML project directory and edit it there (see section Calling the Build Process from the Command Line).

1.5.3 Calling the Build Process from the Command Line

You can execute the Ant build process from your eLML project folder to generate the output formats. To provide the necessary parameters, copy the file `$PUBLISH_USECASE/ant/params.properties` to your project folder and edit the file to update the parameter values. To execute the build process, use the following command (replace `$PUBLISH_USECASE` with the path to the publish usecase directory):

```
ant -Dbasedir=. -f $PUBLISH_USECASE/ant/build.xml generate-output
```

Be careful to provide the `-Dbasedir=.` parameter, otherwise your generated files will end up in the directory where the buildfile is located.